

Gus's Workboard: Grammar Contraption Parts

A list of silly, physics-style, plug-and-play machine parts for Gus's Workboard. Each part does something funny on the board and has a real grammar job hidden inside it. Prepared by Claudia (game and instructional design review), 2026-10-07.

Teacher's request: "claudia, lets develop a list of features that can be physics/contraption/rube goldberg fun plug and play machine elements that are rooted in grammatical concepts"

Teacher's direction for the whole Workboard: "i want this to be a sandbox style for the kids to play around with grammar in the same STEAM style way they like to play other games, with hidden grammar concepts"

Build tiers:

- **NOW:** fits the current grammar engine.
- **SOON:** needs a small engine addition.
- **LATER:** needs grammar the engine does not teach yet.

Rules every part follows

- **Wrong is private.** A wrong placement makes the part wobble, clunk or jam on the student's own screen only. Nobody else sees it.
- **Effort pays.** A fixed jam still earns the full run and the movie.
- **No gambling.** There are no spinning prize wheels, loot boxes or random rewards. The grammar causes every reaction.
- **iPad first.** Every part snaps on by tap or drag. Targets are 44px or larger, and nothing needs hover.
- **Set per student.** The teacher can turn parts on or off.
- **Silly is welcome.** Sounds, bounces, sparks and steam are all good.

Already on the Workboard (shipped 2026-10-07)

Every fun part now has a grammar job:

Part	What it does	Hidden grammar
Big Horn (new)	HONK! Blasts at the end	Exclamation point, or the "!" right after an interjection
Spring Mat	BOING! Bounces in a word	Adjective (describing word)
Fan	WHOOSH! Blows in a word	Adverb (how word)
Ramp	WHEEE! Rolls in a word	Preposition (where word)
Conveyor Belt	CLANK-CLANK, carries a word	Conjunction (joining word)
Bucket Drop	SPLOSH! Drops in a word	Article (a, an, the)

Part	What it does	Hidden grammar
Pulley	HEAVE-HO! Hoists the next word	Capital letter
Bell	DING!	Period
Dominoes	CLACK-CLACK	Comma
Duplicator (new)	COPY! COPY! Copies the next noun	Plural noun, and the verb has to agree

1. Capital letters and punctuation

1. Proper Name Stamp (SOON)

- **What it does:** A giant rubber stamp hangs over the pipe. When a name passes, it slams down and a gold capital letter pops up with a clang.
- **Grammar job:** Names of people, places and days start with a capital letter.
- **Trains:** Proper nouns.
- **Example:** "gus visited vermont" becomes "Gus visited Vermont."

2. Mood Meter Valve (NOW)

- **What it does:** A pressure gauge whose needle swings from "calm" to "BIG FEELING." It opens the Bell or the Big Horn.
- **Grammar job:** Matches the end punctuation to the sentence. Calm gets a period and a strong feeling gets an exclamation point.
- **Trains:** Choosing end punctuation on purpose.
- **Example:** "The roof is on fire." becomes "The roof is on fire!"

3. Comma List Train (SOON)

- **What it does:** A little train with one cart per item. Each coupler between carts is a comma, and the last cart hooks on with an "and" hook.
- **Grammar job:** Commas between items in a list.
- **Trains:** Commas in a series.
- **Example:** "I packed pizza chips and cake" becomes "I packed pizza, chips, and cake."

4. Sentence Sprinter Flag (NOW)

- **What it does:** A starting gun fires at the first word and a checkered flag drops at the last mark.
- **Grammar job:** A capital letter at the start and punctuation at the end.
- **Trains:** Sentence boundaries.
- **Example:** "the dog barked" gets a Pulley and a Bell, then the flag waves.

2. Nouns and articles

5. Irregular Trapdoor (SOON)

- **What it does:** The Duplicator's weird cousin. The marble drops through a trapdoor and comes out odd: a cartoon "mouses" gets crushed and "mice" comes out.
- **Grammar job:** Irregular plurals.
- **Trains:** Irregular plural nouns.
- **Example:** "two mouse" becomes "two mice."

6. A/An Vowel Sniffer (NOW)

- **What it does:** A vacuum nozzle on a spring sniffs the next word's first letter. A vowel sound makes it sneeze and swap the Bucket Drop's "a" for "an."
- **Grammar job:** a versus an.
- **Trains:** Articles before vowel sounds.
- **Example:** "a apple" becomes "an apple."

3. Pronouns

7. Pronoun Teleporter (NOW)

- **What it does:** A glowing yellow pad. When a noun repeats, the second one steps on it and zaps into a pronoun.
- **Grammar job:** Replaces a repeated noun with the right pronoun.
- **Trains:** Pronoun reference.
- **Example:** "Mia fed the cat. Mia smiled." becomes "Mia fed the cat. She smiled."

8. Subject and Object Turnstile (SOON)

- **What it does:** A two-way gate. The doer side lets I, he and she through. The receiver side lets me, him and her through. A wrong pronoun makes the gate spin and bonk.
- **Grammar job:** Pronoun case.
- **Trains:** Subject pronouns and object pronouns.
- **Example:** "Him threw the ball to I" becomes "He threw the ball to me."

9. Possessive Tag Gun (LATER)

- **What it does:** A sticker gun shoots a name tag onto an item.
- **Grammar job:** An apostrophe and "s" show who owns something.
- **Trains:** Possessives.
- **Example:** "the dogs bone" becomes "the dog's bone."

4. Verbs and time

10. Agreement Gears (NOW)

- **What it does:** The noun machine and the action machine each get a gear. Matching gears mesh with a smooth whir. A mismatch grinds and throws sparks.
- **Grammar job:** Subject and verb agreement.
- **Trains:** One and more than one.
- **Example:** "The dogs runs" becomes "The dogs run."

11. Time Tunnel (NOW)

- **What it does:** A three-lane tunnel with a past, present and future gate. The action machine changes shape as the marble passes through.
- **Grammar job:** The verb's time matches the Clock and any time word.
- **Trains:** Consistent verb tense.
- **Example:** "Last night, we walk" becomes "Last night, we walked."

12. Irregular Past Press (SOON)

- **What it does:** A heavy press. "goed" goes in and "went" gets stamped out.
- **Grammar job:** True irregular past forms.
- **Trains:** Irregular past verbs.
- **Example:** "He runned home" becomes "He ran home."

5. Adjectives and adverbs

13. Describe Sorter Chute (NOW)

- **What it does:** A chute with slots in order (opinion, size, color). The words roll in and clack into their slots.
- **Grammar job:** Describing words in the right order.
- **Trains:** Adjective order.
- **Example:** "a red big ugly truck" becomes "an ugly big red truck."

14. How-Dial (NOW)

- **What it does:** A big dial on the Fan. "slowly" makes the marble creep and "quickly" makes it zoom.
- **Grammar job:** How the action happens, with the adverb beside its verb.
- **Trains:** Adverbs of manner.
- **Example:** "The turtle moved slowly."

15. Opener Slingshot (NOW)

- **What it does:** A slingshot flings a how word to the front of the line, and a tiny comma domino tips behind it.
- **Grammar job:** An adverb opener with its comma.
- **Trains:** Sentence openers.
- **Example:** "Suddenly, the lights went out."

16. Size-Up Inflator (LATER)

- **What it does:** A bike pump. Each pump grows the describing word's balloon: tall, taller, tallest.
- **Grammar job:** Comparing forms.
- **Trains:** Comparatives and superlatives.
- **Example:** "Gus is the tall robot" becomes "Gus is the tallest robot."

6. Prepositions

17. Where-To Switchyard (NOW)

- **What it does:** A rail switch with three paths. The marble goes over a bridge, under a tunnel or through a ring, depending on the preposition.
- **Grammar job:** The preposition shows where.
- **Trains:** Position words.
- **Example:** "The cat jumped over the fence."

18. Dead-End Detector (NOW)

- **What it does:** A Ramp with no landing gets a "ROAD CLOSED" sign. Add a noun and the landing pad pops into place.
- **Grammar job:** Every preposition needs a noun after it.
- **Trains:** Complete prepositional phrases.
- **Example:** "The cat sat on." becomes "The cat sat on the mat."

7. Conjunctions

19. Merge Funnel (NOW)

- **What it does:** A big purple funnel. Two noun marbles pour in and one fat marble comes out.
- **Grammar job:** Two nouns joined by "and" count as more than one.
- **Trains:** Compound subjects.
- **Example:** "Tom and Mia run."

20. Comma Hinge Drawbridge (NOW)

- **What it does:** A drawbridge joins two sentence islands. It only lowers when a comma domino sits at the hinge, right before the joining word.
- **Grammar job:** The comma before and, but or so when two sentences join.
- **Trains:** Compound sentences.
- **Example:** "I was tired, but I kept going."

21. Because Seesaw (LATER)

- **What it does:** A seesaw with a cause on one end and an effect on the other. A "because" weight tips it.
- **Grammar job:** Linking a reason to a result.
- **Trains:** Subordinating conjunctions.
- **Example:** "We stayed in because it rained."

8. Interjections and sentence types

22. Confetti Trapdoor (NOW)

- **What it does:** An orange popper at the start of the pipe. "Wow!" pops up and confetti falls on the first machine.
- **Grammar job:** An interjection goes first, followed by punctuation.
- **Trains:** Interjections.
- **Example:** "Wow! The marble flew."

23. Question Crane (LATER)

- **What it does:** A hook picks up the action machine and swings it in front of the noun.
- **Grammar job:** Flips the word order and adds a question mark.
- **Trains:** Asking questions.
- **Example:** "You are ready." becomes "Are you ready?"

24. Speech Bubble Blower (LATER)

- **What it does:** A bubble pipe. The words inside get quotation marks, and a tag like "she said" follows.
- **Grammar job:** Quotation marks around spoken words.
- **Trains:** Dialogue punctuation.
- **Example:** "I'm done," she said.

Chain reactions (combos)

- **Plural Pileup.** The Duplicator copies the noun, then the Agreement Gears grind until the verb changes to match: "one dog barks" becomes "two dogs bark."

- **Time Warp.** The Clock set to the future, the Time Tunnel and a time word like "Next" all have to agree. When they do, the movie plays.
- **Dramatic Entrance.** The Confetti Trapdoor, the Mood Meter Valve and the Big Horn: "Wow" opens the line, the gauge reads high, and the Horn blasts the exclamation point.
- **Big Merge.** The Merge Funnel, the Agreement Gears and the Comma Hinge Drawbridge: two nouns pour in, the verb gear shifts to plural, and the bridge lowers to join a second sentence.
- **Road Closed Repair.** The Dead-End Detector and the Where-To Switchyard: add the missing noun, the landing appears, and the marble goes over, under or through on the TV.

Suggested build order

1. **First NOW batch:** Agreement Gears, Time Tunnel, Merge Funnel and the A/An Vowel Sniffer. These pay off the most and each shows grammar through motion.
2. **Second NOW batch:** the rest of the NOW parts.
3. **SOON batch:** Proper Name Stamp, Comma List Train, the Subject and Object Turnstile, and the two irregular parts.
4. **LATER:** after the engine teaches questions, possessives and quotation marks.